

```

clc;
clear;
close all;

%% DIGITAL SIGNAL PROCESSING ANALYSIS
% Sampling, FFT, Decimation, Interpolation and FIR Filtering

%% 1. Signal Generation

Fs = 1000;           % Sampling frequency (Hz)
T = 1;               % Signal duration (seconds)

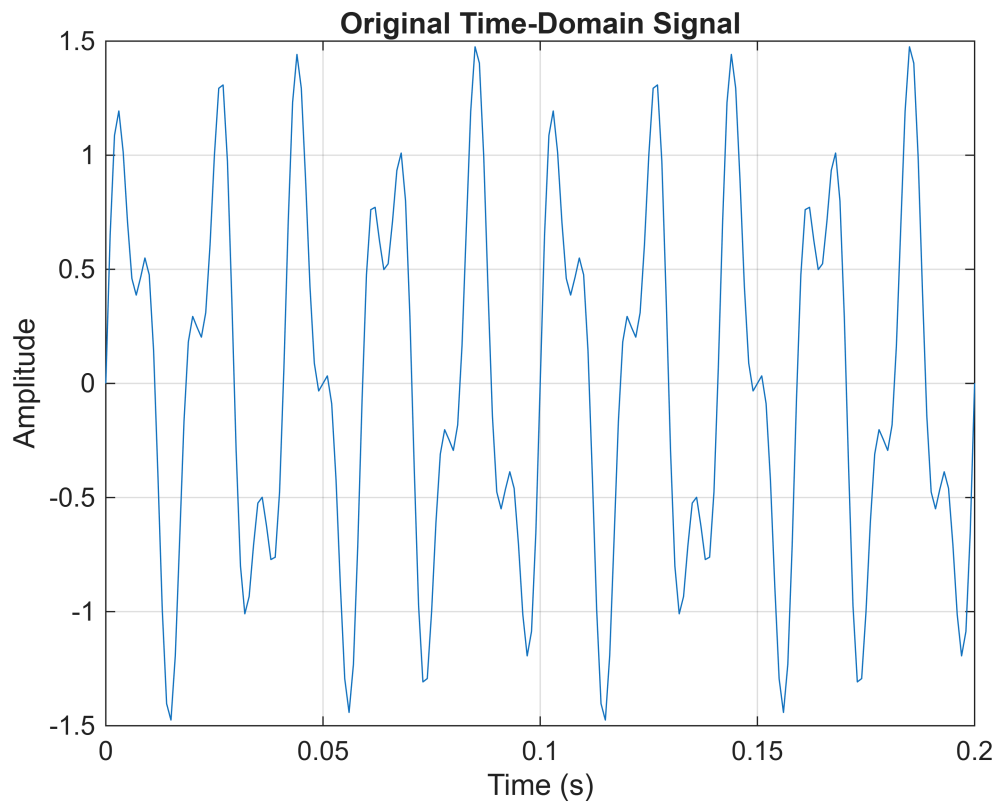
t = 0:1/Fs:T-1/Fs;

f1 = 50;              % First frequency component
f2 = 120;             % Second frequency component

x = sin(2*pi*f1*t) + 0.5*sin(2*pi*f2*t);

% Time-domain signal
figure;
plot(t, x);
xlabel('Time (s)');
ylabel('Amplitude');
title('Original Time-Domain Signal');
grid on;
xlim([0 0.2]);

```



%% 2. FFT - Frequency Analysis

```
N = length(x);
```

```
X = fft(x);
```

```
frequency = (0:N-1)*(Fs/N);
```

```
magnitude = abs(X)/N;
```

```
figure;
```

```
plot(frequency, magnitude);
```

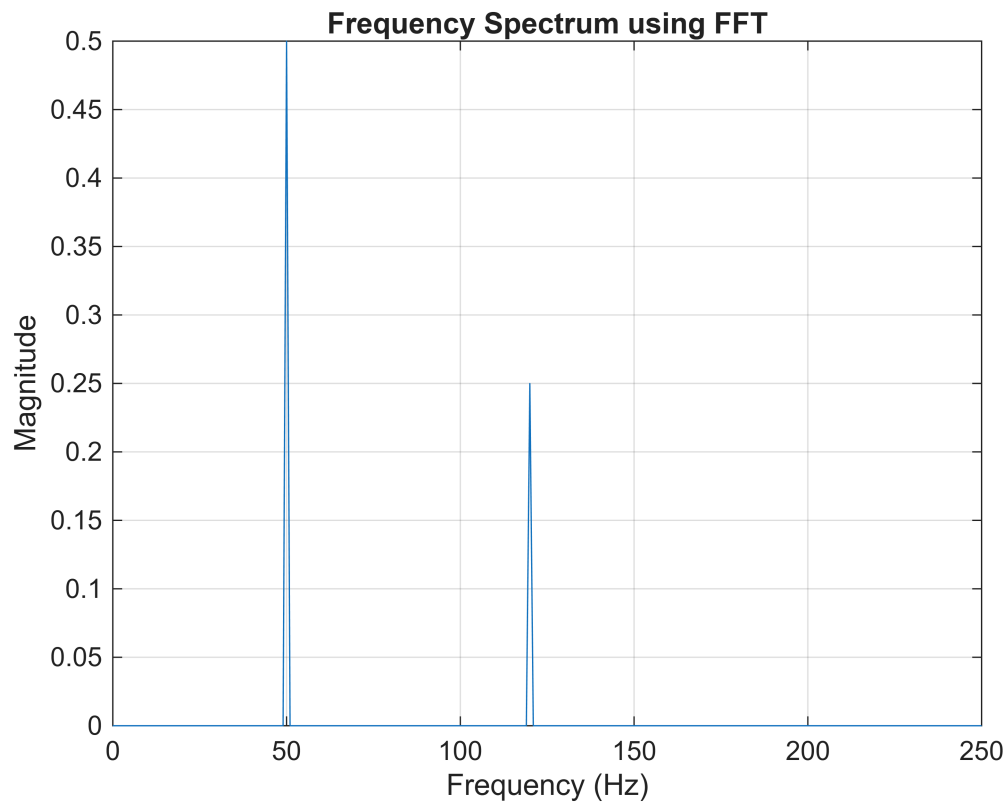
```
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude');
```

```
title('Frequency Spectrum using FFT');
```

```
grid on;
```

```
xlim([0 250]);
```



%% 3. Decimation (Downsampling)

```
D = 2; % Decimation factor
```

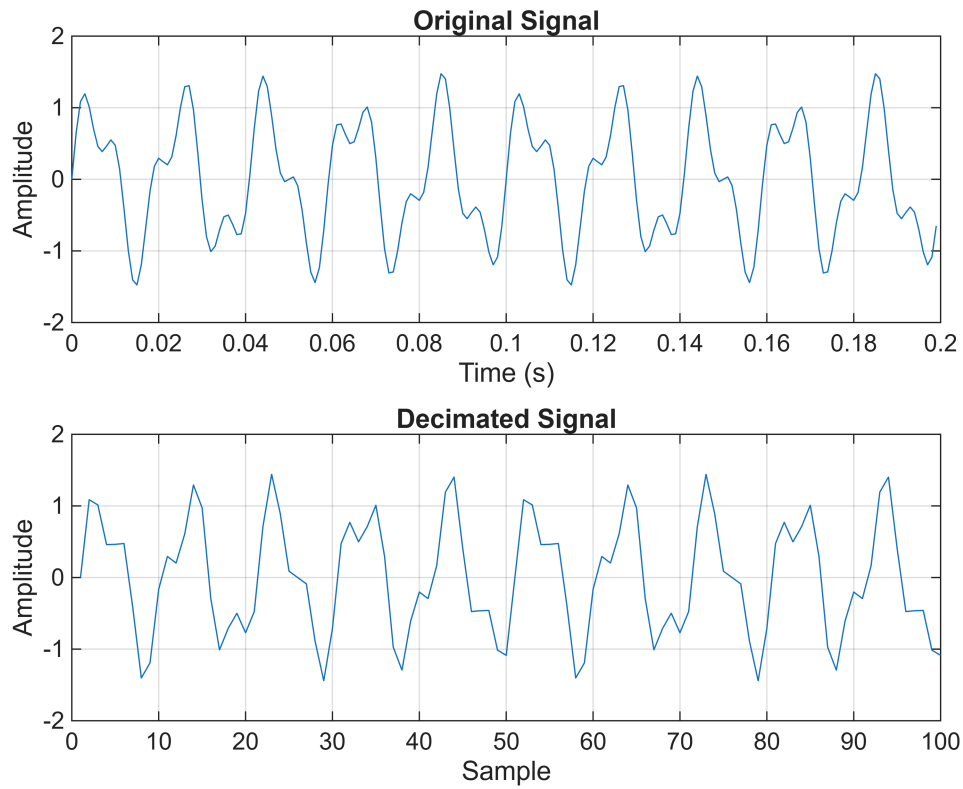
```
x_decimated = x(1:D:end);
```

```
Fs_decimated = Fs/D;
```

```
figure;
```

```
subplot(2,1,1);
plot(t(1:200), x(1:200));
xlabel('Time (s)');
ylabel('Amplitude');
title('Original Signal');
grid on;
```

```
subplot(2,1,2);
plot(x_decimated(1:100));
xlabel('Sample');
ylabel('Amplitude');
title('Decimated Signal');
grid on;
```



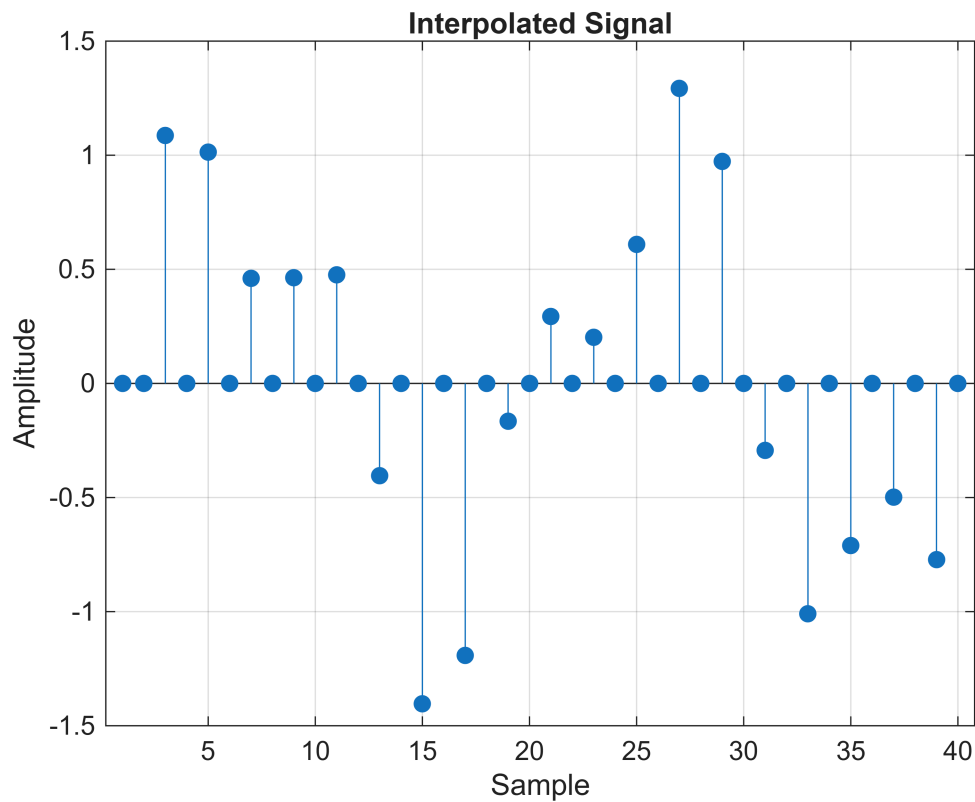
%% 4. Interpolation (Upsampling)

```
L = 2; % Interpolation factor

x_interpolated = zeros(1, length(x_decimated)*L);

x_interpolated(1:L:end) = x_decimated;

figure;
stem(x_interpolated(1:40), 'filled');
xlabel('Sample');
ylabel('Amplitude');
title('Interpolated Signal');
grid on;
```



```
%% 5. FIR Low-Pass Filter
```

```
% Moving-average FIR filter
```

```
filter_length = 11;
```

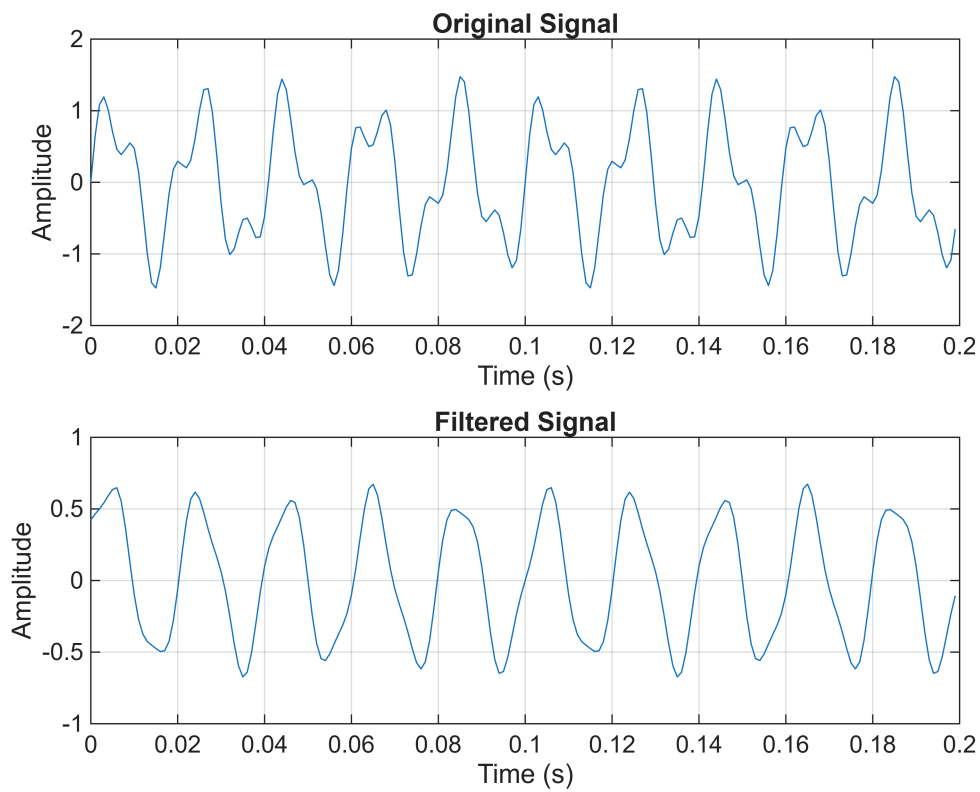
```
h = ones(1, filter_length) / filter_length;
```

```
x_filtered = conv(x, h, 'same');
```

```
figure;
```

```
subplot(2,1,1);
plot(t(1:200), x(1:200));
xlabel('Time (s)');
ylabel('Amplitude');
title('Original Signal');
grid on;
```

```
subplot(2,1,2);
plot(t(1:200), x_filtered(1:200));
xlabel('Time (s)');
ylabel('Amplitude');
title('Filtered Signal');
grid on;
```



```
%% 6. FFT of Filtered Signal
```

```
X_filtered = fft(x_filtered);
```

```
filtered_magnitude = abs(X_filtered)/N;
```

```
figure;
```

```
plot(frequency, filtered_magnitude);
```

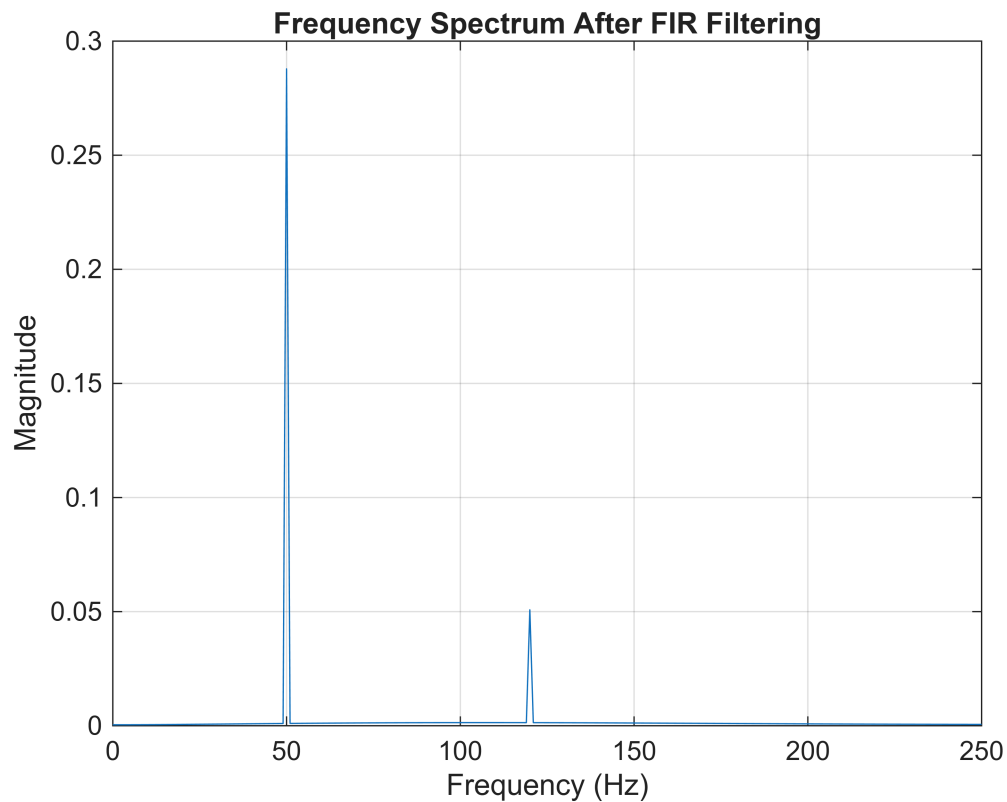
```
xlabel('Frequency (Hz)');
```

```
ylabel('Magnitude');
```

```
title('Frequency Spectrum After FIR Filtering');
```

```
grid on;
```

```
xlim([0 250]);
```



```
%% 7. Summary
```

```
disp('-----');
```

```
-----
```

```
disp(' DIGITAL SIGNAL PROCESSING ANALYSIS');
```

```
DIGITAL SIGNAL PROCESSING ANALYSIS
```

```
disp('-----');
```

```
-----
```

```
disp(['Sampling Frequency: ', num2str(Fs), ' Hz']);
```

```
Sampling Frequency: 1000 Hz
```

```
disp(['Signal Frequencies: ', num2str(f1), ' Hz and ', num2str(f2), ' Hz']);
```

```
Signal Frequencies: 50 Hz and 120 Hz
```

```
disp(['Decimation Factor: ', num2str(D)]);
```

```
Decimation Factor: 2
```

```
disp(['Interpolation Factor: ', num2str(L)]);
```

Interpolation Factor: 2

```
disp(['FIR Filter Length: ', num2str(filter_length)]);
```

FIR Filter Length: 11

```
disp('FFT, decimation, interpolation and filtering completed.');
```

FFT, decimation, interpolation and filtering completed.